

PathFinder — Product Requirements Document (PRD)

1. Product Vision

PathFinder envisions a world where every student has access to the right opportunities at the right time, matched intelligently to their unique skills, interests, and academic background. The platform eliminates the noise and fragmentation of traditional opportunity discovery by providing a centralized, AI-enhanced ecosystem.

2. Product Goals

- Centralize Opportunity Discovery:** Aggregate internships, scholarships, grants, and hackathons into one searchable platform.
 - Intelligent Matching:** Automatically score how well a student's profile aligns with each opportunity's requirements.
 - Streamline Applications:** Reduce the time students spend on application documents through AI generation.
 - Enable Transparent Tracking:** Give students real-time visibility into their application status.
 - Empower Organizations:** Provide tools for efficient applicant review and management.
 - Ensure Platform Governance:** Give administrators comprehensive oversight and control.
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3. Business Goals

- Achieve a functional MVP deployment within 4 weeks.
 - Ensure the platform can operate within free-tier cloud hosting limits.
 - Build extensible architecture that supports future feature development without major refactoring.
 - Create documentation comprehensive enough that any developer can onboard within 1 day.
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4. User Personas

Persona 1: Aarav (Student)

- Age:** 21 | **Year:** 3rd Year B.Tech | **Major:** Computer Science
- Goals:** Find relevant internships, build a strong profile, generate tailored application documents.
- Frustrations:** Spends hours searching different websites; writes generic cover letters because he lacks time.
- How PathFinder helps:** Smart matching shows him the most relevant opportunities; AI generates personalized SOPs and cover letters.

Persona 2: Priya (Organization Representative)

- Age:** 35 | **Role:** HR Manager at a tech startup
- Goals:** Find qualified interns quickly, reduce time reviewing unqualified applications.
- Frustrations:** Receives 200+ applications per posting; no way to assess candidate fit automatically.
- How PathFinder helps:** AI match scores rank applicants by suitability; candidate summaries provide instant profile overviews.

Persona 3: Admin (Platform Administrator)

- Age:** 28 | **Role:** Platform operations manager
 - Goals:** Monitor platform health, manage user issues, ensure data integrity.
 - Frustrations:** No visibility into platform usage; user complaints arrive via email with no tracking.
 - How PathFinder helps:** Analytics dashboard shows real-time metrics; support ticketing system tracks every user issue.
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5. User Journey

Student Journey

- Register** → Create account with email, password, and role selection.
- Build Profile** → Add personal details, skills, certifications, portfolio links, and upload resume.
- Discover Opportunities** → Browse the dashboard, use search and filters, or view AI recommendations.
- Evaluate Fit** → View match score for a specific opportunity before applying.
- Apply** → Submit application with optional cover letter; system calculates match score.
- Generate Documents** → Use AI assistant to create SOP, cover letter, or essay for the opportunity.
- Track Applications** → Monitor status changes (Pending → Reviewing → Shortlisted → Accepted/Rejected).

- 8. **Receive Notifications** → Get notified when application status changes.
- 9. **Get Support** → Submit a support ticket if issues arise.

Organization Journey

- 1. **Register** → Create account as Organization with company name.
- 2. **Set Up Profile** → Add company description, logo, website, and industry.
- 3. **Post Opportunity** → Create a new internship, scholarship, grant, or hackathon listing.
- 4. **Review Applicants** → View applications sorted by match score.
- 5. **Assess Candidates** → View AI-generated candidate summaries, download resumes.
- 6. **Update Statuses** → Accept, reject, or shortlist applicants individually or in batch.
- 7. **Manage Listings** → Archive or delete old opportunities.

Admin Journey

- 1. **Login** → Access the admin dashboard with elevated privileges.
- 2. **View Analytics** → Review platform-wide metrics (users, opportunities, applications, status breakdowns).
- 3. **Manage Users** → Search, filter, change roles, or delete user accounts.
- 4. **Manage Opportunities** → View and remove inappropriate or expired listings.
- 5. **Handle Support Tickets** → Review, respond to, and resolve user support requests.
- 6. **Monitor System Health** → Check server uptime, memory usage, and database statistics.
- 7. **Export Data** → Download analytics as CSV for external reporting.
- 8. **Review Audit Log** → Track all significant platform actions for compliance.

6. Feature List

Core Features

Feature	Description	Priority
User Registration & Login	Email/password auth with JWT tokens	P0
Role-Based Access Control	Three roles: STUDENT, ORGANIZATION, ADMIN	P0
Student Profile Management	Full profile with skills, certs, links, resume	P0
Organization Profile Management	Company details, logo, website	P0
Opportunity CRUD	Create, read, update, delete opportunity listings	P0
Application Submission	Students apply with cover letter	P0
Application Status Management	Organizations update statuses (6 states)	P0
AI Match Scoring	0-100 compatibility score per application	P0
Search & Filter	Category, location, keyword search	P0

Enhanced Features

Feature	Description	Priority
AI Document Generation	SOP, Cover Letter, Essay, Personal Statement	P1
Resume PDF Parsing	Extract profile data from uploaded resume	P1
Profile Strength Analysis	Completeness score with actionable suggestions	P1
AI Recommendations	Personalized opportunity suggestions	P1
Notification System	In-app notifications for status changes	P1
Bookmarking	Save opportunities for later review	P1
Application Timeline	Chronological event history per application	P1

Administrative Features

Feature	Description	Priority
Admin Analytics Dashboard	Platform-wide metrics and breakdowns	P1
User Management	Search, filter, role changes, account deletion	P1
Opportunity Management	Admin oversight of all listings	P1
Audit Logging	Track all significant platform actions	P1
System Health Monitor	Server uptime, memory, database stats	P1
Data Export (CSV)	Export analytics data for external use	P2
Support Ticketing System	Two-way communication with threaded messages	P1

Security Features

Feature	Description	Priority
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Password Hashing	bcrypt with 10 salt rounds	P0
JWT Authentication	24-hour token expiry	P0
Rate Limiting	5 requests per 15 minutes on auth routes	P0
File Upload Validation	5MB limit, allowed extensions only	P0
Password Reset Flow	Secure token-based email reset	P1
Protected File Access	Documents require authentication, images are public	P1

7. MVP Scope

The Minimum Viable Product includes all P0 features plus the core P1 features (AI Match Scoring, Search, Notifications). The MVP demonstrates the complete end-to-end workflow for all three user roles.

8. Future Features

- Mobile application (React Native)
- Real-time chat between students and organizations
- Calendar integration for deadline tracking
- Email campaign automation for organizations
- Multi-language support (internationalization)
- Government opportunity API integration
- AI interview preparation module
- Recommendation letter generator

9. Success Metrics

Metric	Target
Platform Uptime	99%+ (within free tier constraints)
Application-to-Opportunity Conversion	>30% of viewed opportunities receive applications
Match Score Accuracy	Students report match scores as "helpful" in >70% of cases
Document Generation Usage	>50% of active students use at least one AI generation tool
Time to First Application	<10 minutes from registration to first application

10. Acceptance Criteria

1. A new student can register, build a profile, and submit an application in under 10 minutes.
2. An organization can post an opportunity and review applicants with match scores.
3. An admin can view analytics, manage users, and resolve support tickets.
4. All API endpoints return appropriate HTTP status codes and error messages.
5. The platform is accessible via public Vercel and Render URLs.
6. Dark mode and light mode toggle functions correctly across all pages.